Supervised learning

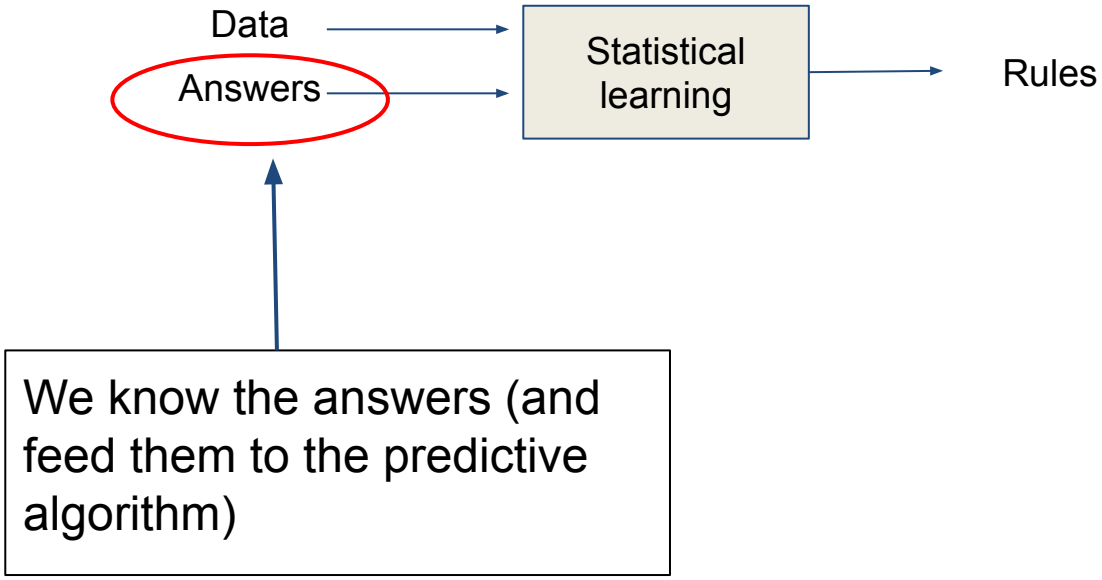
Train the learners

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Supervised learning problems





Why supervised?

Training examples

measured variables / **features** on *n* examples

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 0.12 & 1.5 & \dots & 0.9 \\ 2.05 & 0.95 & \dots & 1.1 \\ \vdots & \vdots & \ddots & \vdots \\ 3.5 & 0.88 & \dots & 1.75 \end{bmatrix}$$

labels / **target variables** (e.g. phenotypes) on n examples



Unsupervised learning

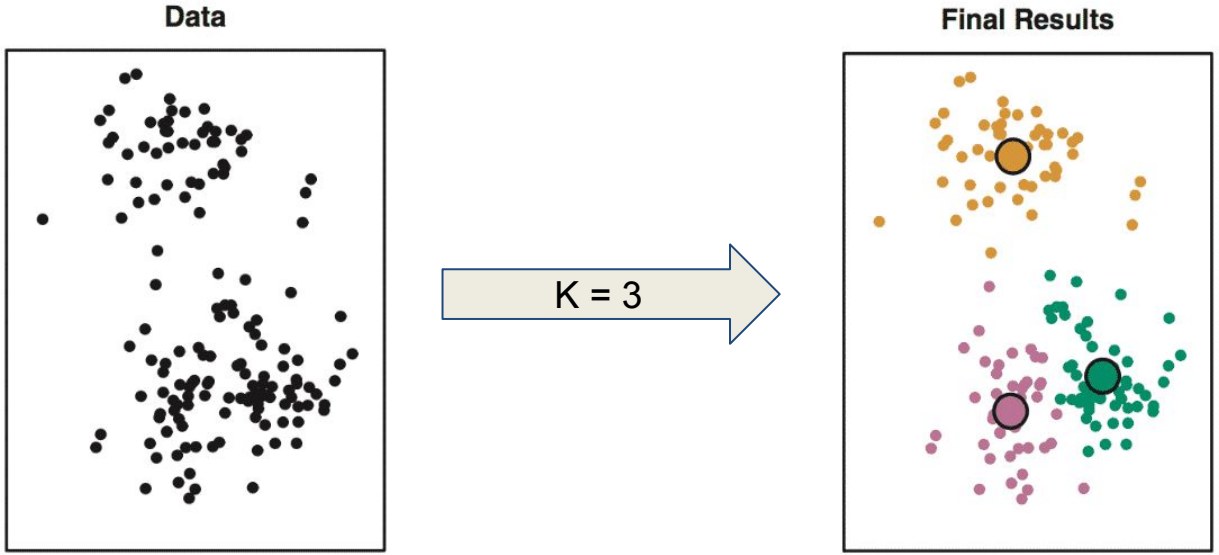
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Unsupervised example: K-means clustering





source:
https://www.iotforall.com/machine-learning-crash-course-unsupervised-learning

Supervised learning: Regression and classification



Supervised learning problems

- Regression (**predictive**) problems
- Classification (**predictive**) problems

with deep learning also:

- Computer vision
- Image segmentation / object detection / counting
- Sound recognition
- Forecasting
- Generative modelling
- NLP: natural language processing



Supervised learning problems

- Regression (**predictive**) problems
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<u>with deep learning also</u>:

- Computer vision
- Image segmentation / object detection / **counting**
- Sound recognition
- Forecasting
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- NLP: natural language processing

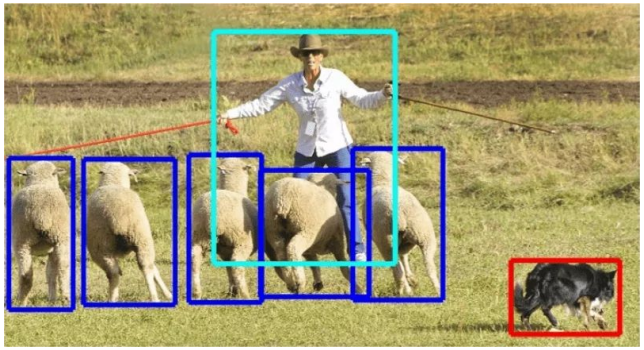
- **microbiology** (e.g. counting bacterial colonies)
- **surveillance** / wildlife conservation (e.g. counting people / animals)
- **agriculture** (e.g. counting fruits or vegetables)
- **medicine** (e.g. counting tumor cells in histopathological scans)



Counting objects in images

- **Object Detection-Based Counting**

  image segmentation (drawing bounding boxes) →object detection (what is what) →counting

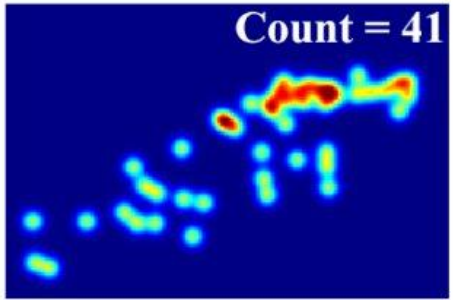


[From: https://neurosys.com/blog/objects-counting-by-estimating-a-density-map]

- **Density Map Estimation**

  for crowded images where individual items overlap (e.g. schools of fish), CNN can predict a continuous density map → sum of pixel values equals the total count





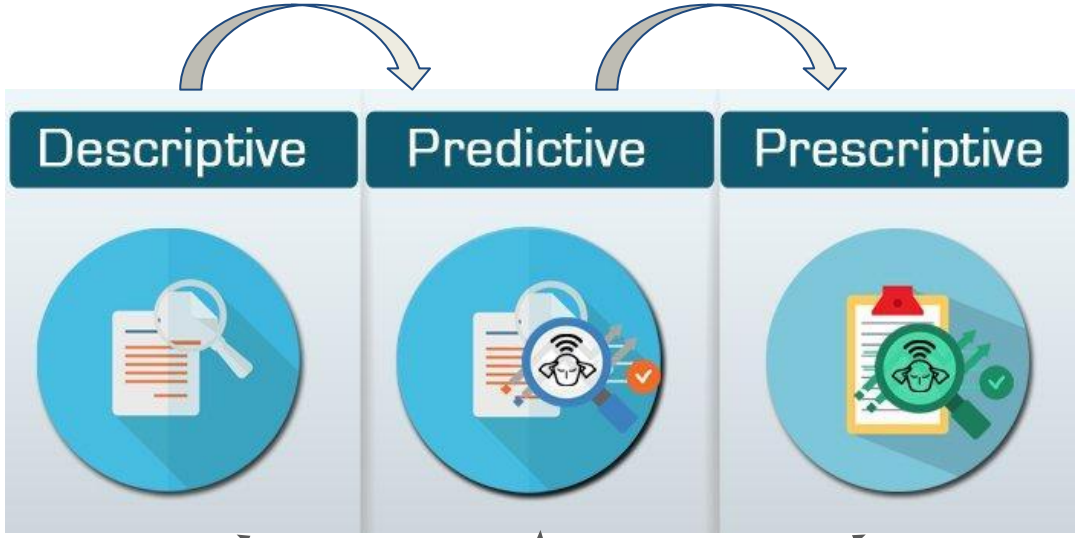
[From: https://ieeexplore.ieee.org/document/7410729]

- **Regression-Based Counting**

  direct regression networks map an entire image patch to a scalar count number without localizing / drawing boxes around single items



Supervised learning (predictive) problems

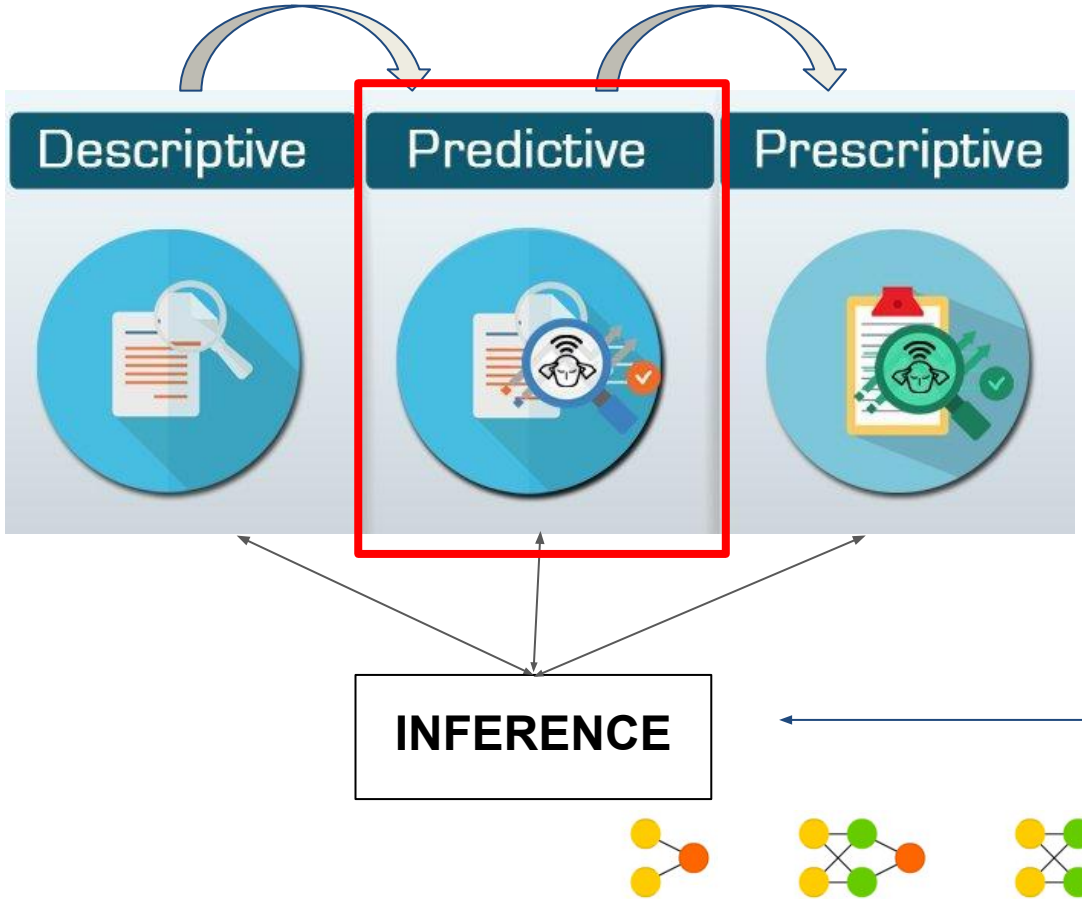


- Know the past
- Predict the future
- Act consequently

INFERENCE



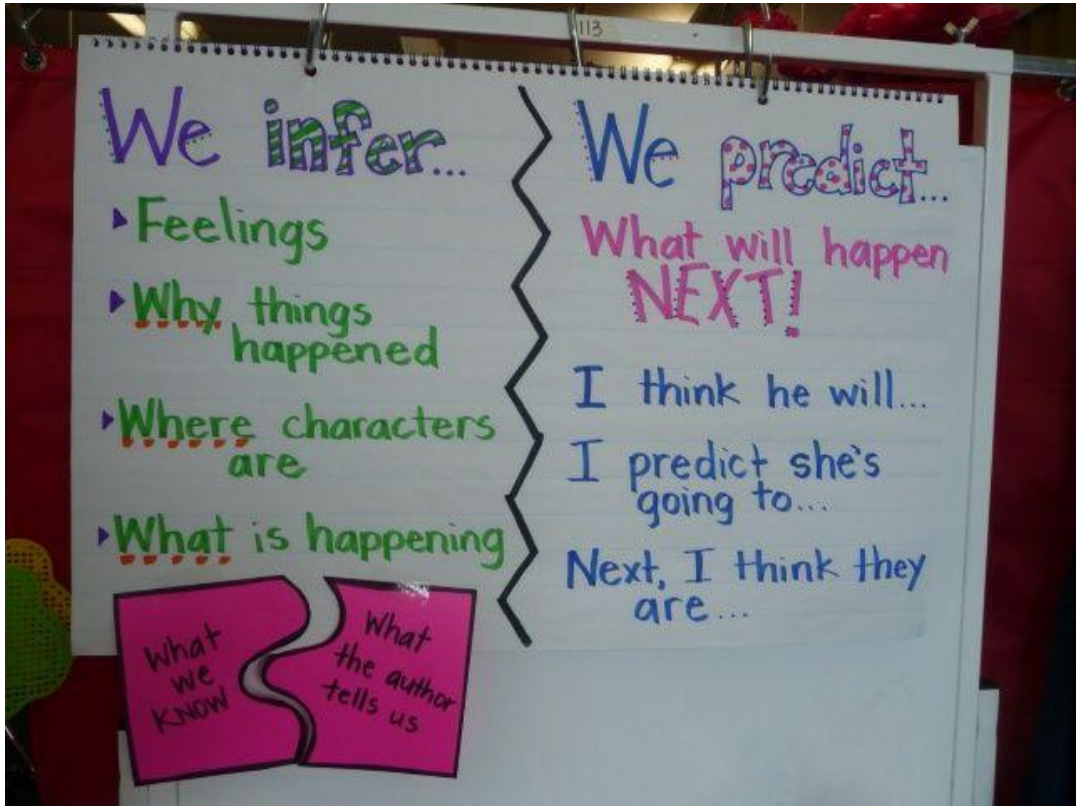
Supervised learning (predictive) problems



- Know the past
- Predict the future
- Act consequently

- A catch-all term
- Can be confusing

Inference vs Prediction



- different statistical problems
- different objectives, different rules … different ballparks
- inference is in general more difficult than prediction

With DL it's usually about predictions!

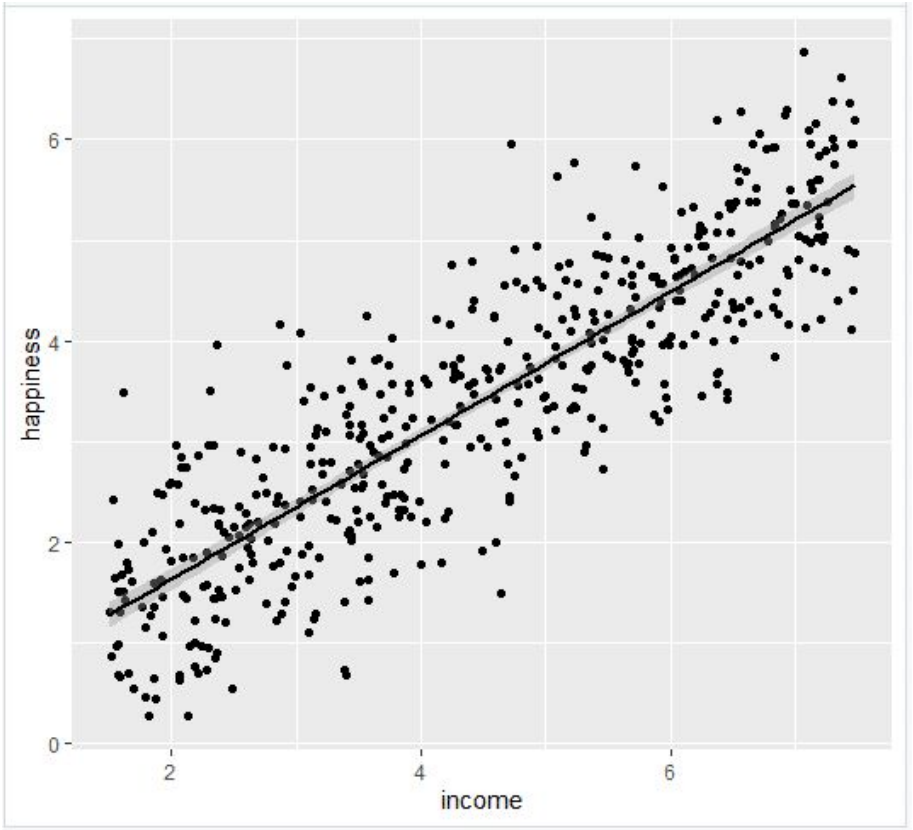


Regression problems

- the response variable **y** is **quantitative**
- e.g.: *height*, *weight*, *yield* (milk, crops), *blood sugar concentration*
- **y** = **target** (dependent) variable (a.k.a. response, objective variable)
- **X** = matrix of **features** (continuous, categorical)
- **regressor**: y = f(x) = **P(X)** ← [predictive machine]



Regression problems - simple regression



happiness = (intercept) + beta*income

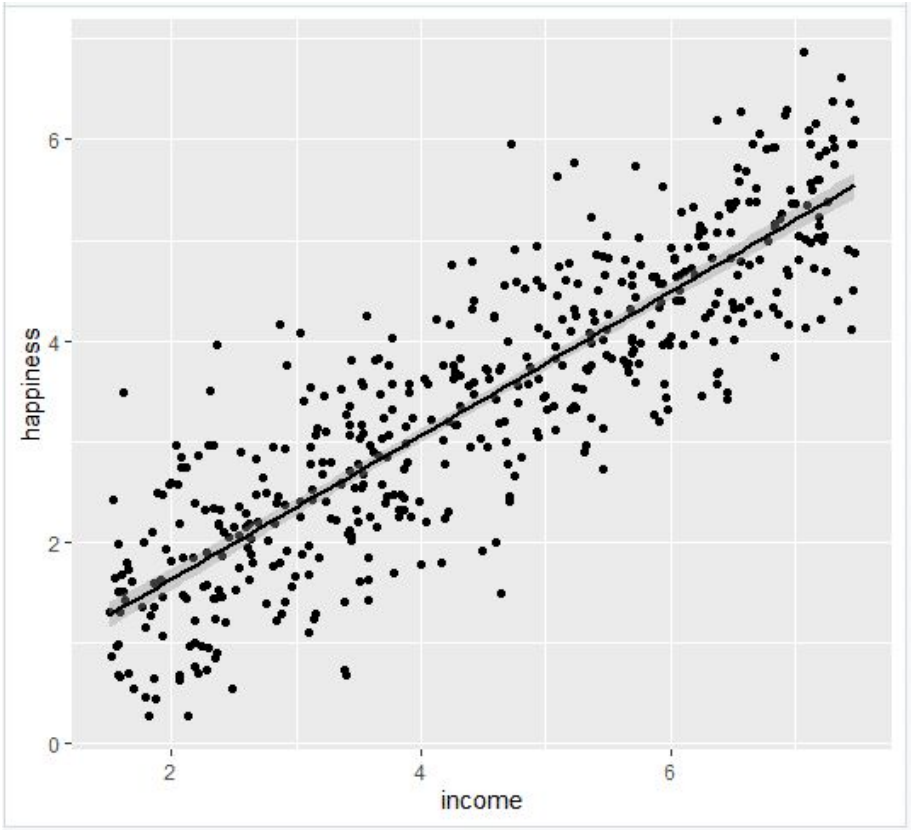
or

income = (intercept) + beta*happiness

Source: https://www.scribbr.com/statistics/linear-regression-in-r/



Regression problems - simple regression



happiness = (intercept) + beta*income

or

income = (intercept) + beta*happiness

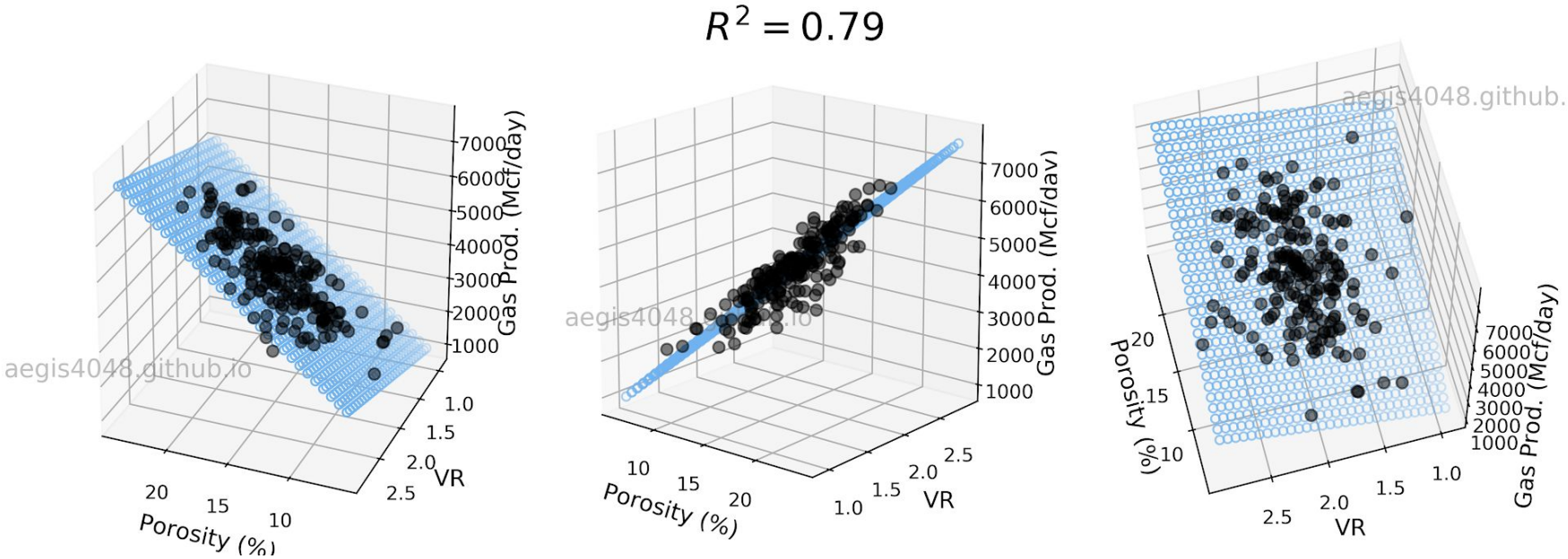
cause → effect?

1. Can we predict backwards?
2. Can we make predictions if there is no cause-effect relationship?

Source: https://www.scribbr.com/statistics/linear-regression-in-r/



Regression problems - multiple regression



- Gas production
- Well porosity %
- Vitrinite reflectance %

Source: https://aegis4048.github.io/mutiple_linear_regression_and_visualization_in_python



Multiple linear regression

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \epsilon$$

- y: target variable
- **β**'s: model coefficients
- X's: features (predictors, independent variables, factors)



Multiple linear regression

$$\mathbf{y} = \beta\mathbf{X} + \mathbf{e}$$

- matrix (compact) notation
- vectors of observations (**y**), coefficients (**β**) and residuals (**e**)
- matrix of features (**X**)



Multiple linear regression

$$\mathbf{y} = \beta\mathbf{X} + \mathbf{e}$$

estimation of coefficients (learning!)

$$\hat{\mathbf{y}} = \hat{\beta}\mathbf{X}$$

→ **predictions!**

- matrix (compact) notation
- vectors of observations (**y**), coefficients (**β**) and residuals (**e**)
- matrix of features (**X**)



Predictions

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_p x_p$$

with the estimated coefficients $\hat{\beta}$ and the feature values **X** we obtain the predicted values $\hat{y}$



Predictions

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_p x_p$$

with the estimated coefficients $\hat{\beta}$ and the feature values **X** we obtain the predicted values $\hat{y}$

→ **how do we find the values for the the model coefficients β?**

(remember: the DL/ML models must learn the rules!)

(think about this: answer later)



There’s more fish in the sea of learning …

- **semi-supervised learning** (only a small fraction of data is labeled)
- **active learning** (labelling is expensive: prioritise which examples -the most informative- should be labeled)
- **self-supervised learning** (identify missing image patches)
- **reinforcement learning** (based on feedback: reward / punishment)
- **transfer learning** (learning is transferred from one model to another)
- **federated learning** (training on distributed devices, no data centralization)
- **online / incremental learning** (continuous updates as new data arrive: e.g. fraud detection)
- **representation learning** (learn data representations, e.g. embeddings rather than direct predictions)
- **generative learning** (create new data - probabilistic models)



Supervised learning recap

- **deep learning** is one of many methods that can be used to solve supervised learning problems
- deep learning is mainly used in **predictive problems**
- (can be used though also for inferential problems and for unsupervised learning)
- we'll see later how to use deep learning to solve linear regression

